# ANDREW ESSAM

**SENIOR SOFTWARE DEVELOPER | REACT & ENGINEERING DESKTOP APPLICATIONS**

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## PROFESSIONAL SUMMARY

Senior Software Developer specializing in React interfaces for engineering and scientific desktop products. Since 2021, has developed production UI for SIDLAB 5, a commercial Windows application used for duct-network acoustics and flow analysis. Delivered four major releases with 1,000+ frontend commits and improvements of up to 60% in application speed. Strengths include interactive diagrams, scientific charts, lab instrument workflows, licensing portals, and modernization of established applications connected to Java and MATLAB-based backends.

## SPECIALIZATIONS

**Engineering UI** | desktop-style workspaces • complex forms • diagram editors • project persistence • scientific plots

**Modernization** | CRA/Webpack to Vite • React upgrades • design-system migration • state refactoring • regression tests

**Commercial products** | license activation • floating/offline licensing • role-based admin portals • instrument workflows

## PROFESSIONAL EXPERIENCE

### Senior Software Developer | SIDLAB | 2021 – Present

Commercial Windows engineering software • React/Java native integration • MATLAB-powered calculation engine (backend maintained by specialist team)

- Owned major parts of the SIDLAB 5 React frontend for a commercial Windows desktop application used to model and analyze 1D duct, exhaust, muffler, ventilation, and acoustics/flow networks; contributed 1,000+ commits across four major releases.
- Built desktop-style navigation and workflows: ribbon controls, sidebars, multi-tab workspaces, editable component properties, multi-value fields, global variables, fluid settings, units, and project save/load behavior.
- Implemented Syncfusion-based network editing with draggable nodes, multi-selection, ports/connectors, copy/paste, alignment, zoom, terminal nodes, rooms, and durable diagram state.
- Built scientific data views and plot management: interactive Syncfusion charts, axis scaling, scientific notation, comparison with measurements, and DOCX report exports with network pictures and plots.
- Delivered lab measurement and acquisition UIs covering rig configuration, calibration, pressure drop, fan speed, absorption, sound transmission loss, NI-DAQ-style hardware options, excitation signals, and live progress charts.

### SIDLAB 5 | Modernization, Integrations & Reliability (continued)

- Modernized the production app from Create React App/Webpack to Vite and toward Redux Toolkit; upgraded React 18, Node 24, and Ant Design 6, replacing Material UI without interrupting release delivery.
- Integrated the frontend with named Java methods through a native window.bridge.call interface; coordinated with specialists responsible for numerical and backend calculations.
- Improved application speed by up to 60% by optimizing network/diagram and chart loading, code splitting, and memoization; fixed production white screens and editing crashes and added approximately 95 Vitest tests for diagram edit rules.

**Related SIDLAB Products | License Administration & Floating Licensing**

- Led the frontend of a React/Ant Design/Redux license-management portal with admin, distributor, and sales roles, customer records, offline license generation, Excel import/export, and version validation.
- Built an Electron/React frontend for floating network-license server administration; also implemented in-product activation, deactivation, borrowing/return, offline licensing, server/IP settings, and feature entitlements.

## SELECTED ADDITIONAL PROJECTS

**SIDLAB website | React, TypeScript, Vite | 2026**

Developed React/TypeScript product-site features including product and download pages, release notes, responsive layouts, and light/dark themes. Site: sidlab.se.

**Household Expenses Tracker | Expo / React Native | 2026 • Independent project**

Built app flows for household budgets, shared contributions, category filtering, Firebase authentication, local SQLite, notifications, localization, and Expo/EAS-based workflows; uses Zustand, React Hook Form, and Zod.

**Pharmacy Inventory App | Flutter / Riverpod | 2026 • Early-stage project**

Implemented an inventory-and-sales app architecture with Drift/SQLite local storage, Firestore sync, barcode scanning, paginated inventory, and multilingual interface.

## TECHNICAL SKILLS

**Core production stack** | React 18, JavaScript, TypeScript, Redux Toolkit, Ant Design 6, Syncfusion Diagrams/Charts, Vite 8, Vitest, CSS Modules

**Desktop & integrations** | Electron, JavaScript-to-Java native bridge, Java-backed controllers, DOCX report generation, hardware/instrument UI

**Other demonstrated tools** | Expo, React Native, Flutter, Riverpod, Firebase, Firestore, SQLite, Zustand, Zod, React Hook Form, Next.js, Prisma

## EDUCATION

Bachelor of Engineering | Benha University | 2020